Stop levels and an earlier profit-take on the SPY gamma book

Experiment report · L/S Gamma Desk, QUANTT

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Question: Do tighter stops (short stop at 0.5x credit, long stop at 0.25x premium) cut the daily 5% expected shortfall of the book by at least 20% at a small cost in P&L (H1)? Does taking profit on shorts at 25% of the credit instead of 50% raise net P&L without a worse tail (H2)?

Method: Pre-registered backtest (hypothesis card committed before any run). The live paper rules (`configs/paper.toml`) are replayed on R2 replay snapshots v1 (config_hash b040760187f0) over the development window 2014-06-02 to 2024-12-31. The hold-out (2025-01-01 to 2026-07-10) was not used. Three variants change only `[exit]`: `tight` (short_stop_mult 0.5, long_stop_frac 0.25; primary for H1), `profit25` (short_profit_frac 0.25; H2) and `both` (all three; reported only). The stop cool-off stays at the live 1 day. Each variant ran once at 1x and 2x spreads and is compared with baseline trial 1 at the same spread multiple. Costs are $0.65 per contract and $0.005 per share, with a $1 minimum per order. Exits are checked once a day at 10:15 ET, as live.

1 Result vs the pass bar

Every check had to hold at both 1x and 2x. ES size is the absolute value of the daily 5% expected shortfall in USD; P&L is net USD for one contract.

H1 (tight): failed. The tail did not shrink; it got slightly worse in both halves and at both spread levels.

Check	1x value	1x bar	2x value	2x bar	Result
ES size, full window	98.1	$\leq$ 77.2	107.1	$\leq$ 83.6	fail
ES size, 2014-06..2019-12	49.0	$<$ 47.9	55.8	$<$ 53.2	fail
ES size, 2020-01..2024-12	134.1	$<$ 132.9	144.4	$<$ 142.9	fail
Net P&L, full window	−2,330.7	$\geq$ −2,359.5	−4,540.3	$\geq$ −4,422.3	fail (2x)

H2 (profit25): failed. It helped in the first half but lost more in the second half, so full-window net P&L was below baseline at both spreads. The tail was unchanged.

Check	1x value	1x bar	2x value	2x bar	Result
Net P&L, full window	−2,153.4	$>$ −2,059.5	−4,170.8	$>$ −4,122.3	fail
Net P&L, 2014-06..2019-12	−922.2	$>$ −986.2	−1,450.7	$>$ −1,511.3	pass
Net P&L, 2020-01..2024-12	−1,231.2	$>$ −1,073.3	−2,720.1	$>$ −2,610.9	fail
ES size, full window	96.6	$\leq$ 106.1	104.5	$\leq$ 115.0	pass

1.1 Full-window summary (no bar for `both`)

Variant	Spread	Net P&L	Sharpe	Max DD	ES 5%	Trades
baseline	1x	−2,059.5	−0.356	3,027	−96.5	131
tight	1x	−2,330.7	−0.398	3,027	−98.1	133
profit25	1x	−2,153.4	−0.373	3,017	−96.6	132
both	1x	−2,555.9	−0.439	3,017	−98.2	134
baseline	2x	−4,122.3	−0.681	4,258	−104.5	131
tight	2x	−4,540.3	−0.732	4,676	−107.1	133
profit25	2x	−4,170.8	−0.691	4,315	−104.5	132
both	2x	−4,722.7	−0.765	4,867	−107.1	134

2 Charts



Figure 1: Stop and profit exits per year, by variant, at 1x and 2x spreads

The tighter stops fired only 3 times in ten and a half years: one in 2015 and two in the 2020 sell-off, at both spread levels (two shorts and one long). The 25% profit-take fired 9 times at 1x and 7 at 2x, mostly in 2020. The baseline never stopped or took profit. With so few events, every result below rests on a handful of positions.

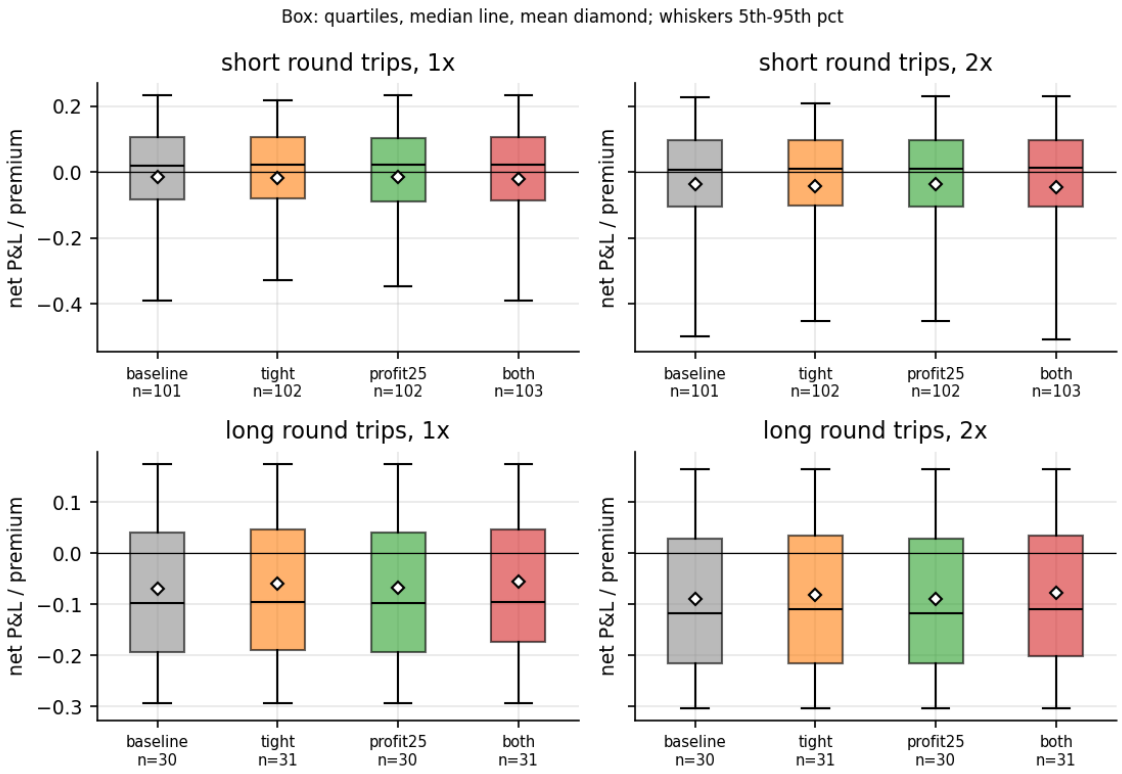


Figure 2: Net P&L of round trips as a multiple of premium, short and long, by variant. Box: quartiles; whiskers: 5th to 95th percentile

The distributions are nearly the same across variants. At 1x the 5th percentile of short round trips is about −0.39x premium for the baseline, `tight` and `profit25` alike (−0.44x for `both`). The whiskers sit on single trades, so their small shifts mean little. The worst short under `tight` is worse than the baseline's (−1.27x against −0.78x premium): it is the 2015-08-24 stop, which closed after the move. Long round trips are almost identical, because only one long was stopped.

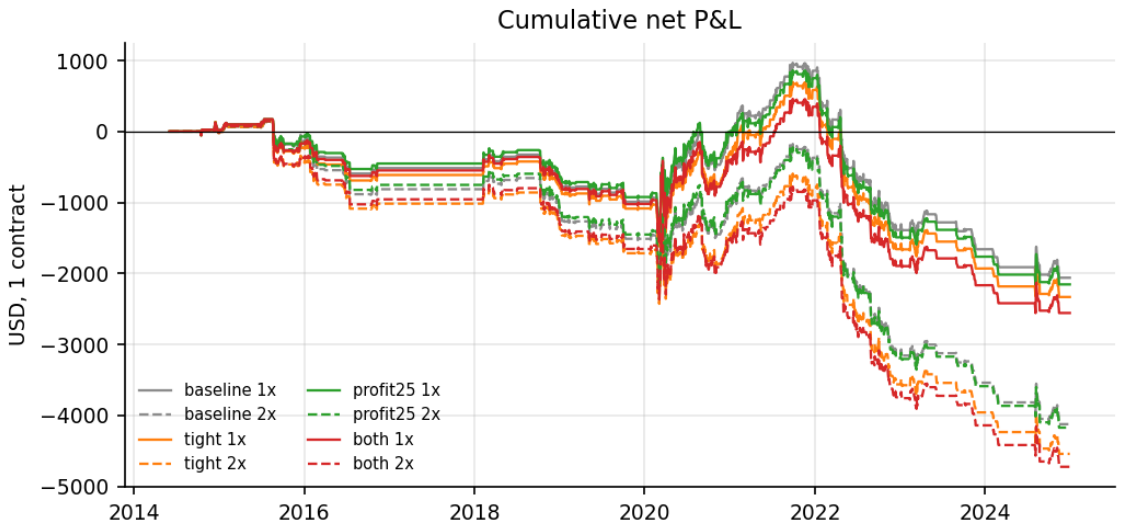


Figure 3: Cumulative net P&L, 1x (solid) and 2x (dashed)

All four curves follow the baseline closely. They separate in two places: August 2015 and the 2020 sell-off, when the stops fired. In both, the stopped variants end up lower. `profit25` runs slightly above the baseline until 2020 and slightly below it afterwards.

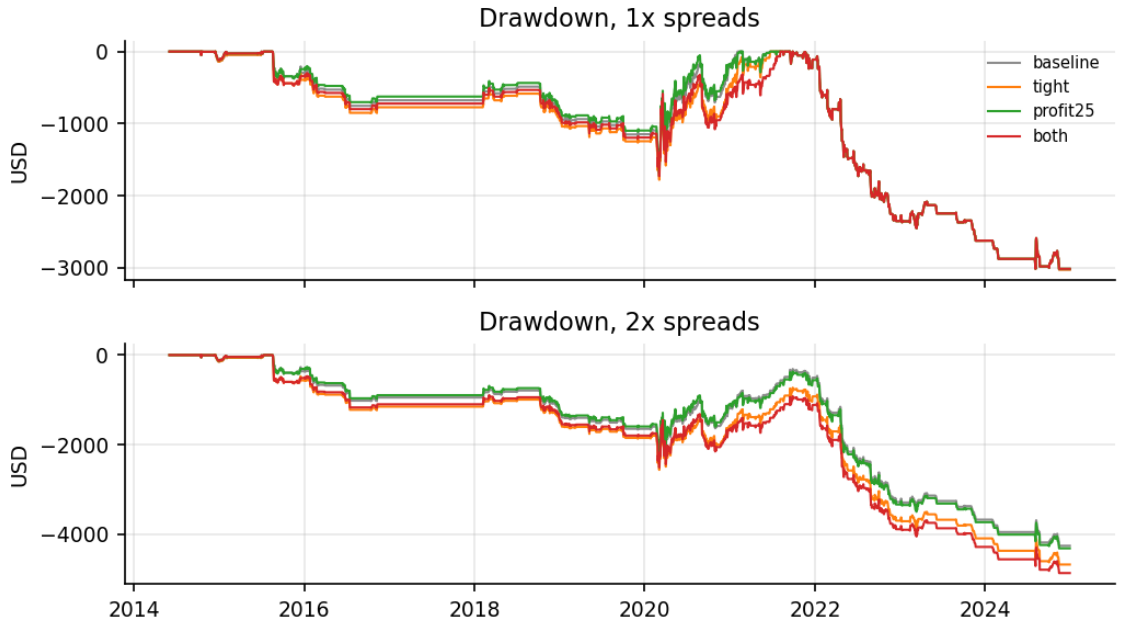


Figure 4: Drawdown from the running peak, 1x and 2x

The 1x maximum drawdown is the 2022-2024 slide for every variant ($3,017 to $3,027). The small gap comes from one profit-take on 2024-10-24 in `profit25` and `both`, which moved their trough to 2024-08-05. At 2x, `tight` and `both` start that slide from a lower level, so their maximum drawdown is $400 to $600 larger.

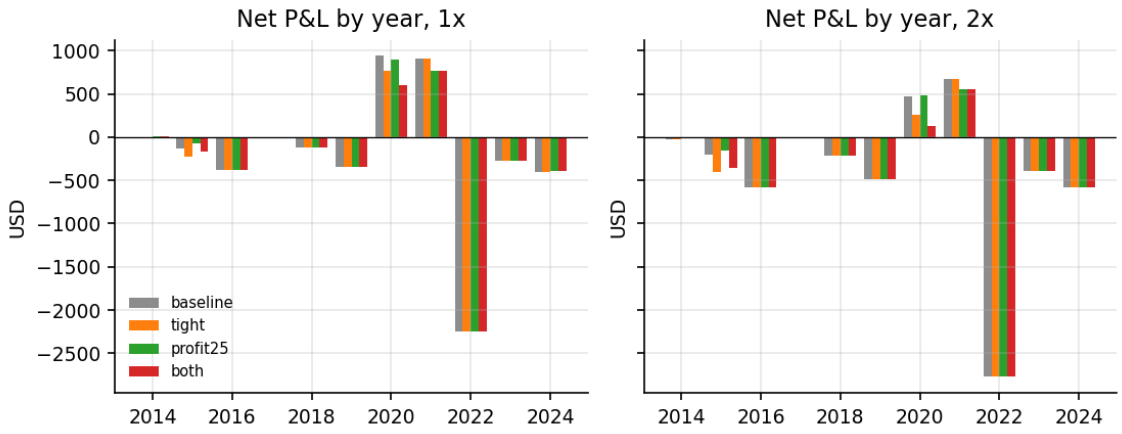


Figure 5: Net P&L by year, 1x and 2x

The yearly bars differ mainly in 2015, 2020 and 2021 (`profit25` and `both` also differ by about $10 in 2014 and 2024). The large 2022 loss is the same for every variant: no exit rule touched it, because no position in 2022 reached the stop levels at the 10:15 check.

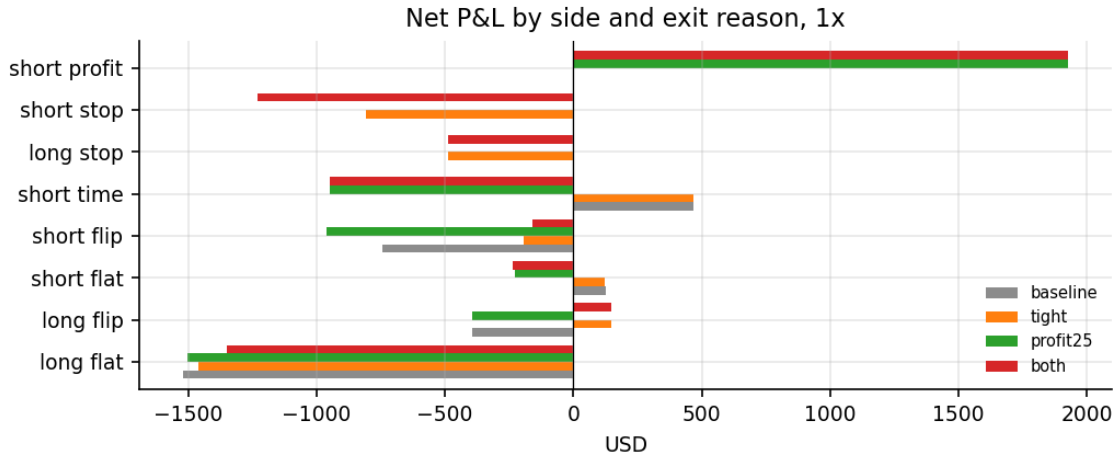


Figure 6: Net P&L by side and exit reason, 1x

The 3 stopped positions lost $1,297 net at 1x. In the baseline, the same 3 positions stayed open and closed on a flip signal for a loss of $924, so the stop locked in losses that later partly recovered. The profit-take works the other way: at 1x, 8 of its 9 positions also exist in the baseline, and on those it earned $254 more than the baseline did, mostly by closing winners before expiry. That gain was more than offset by the shorts opened afterwards. The short time-exit bucket swings from +$470 to −$948, mainly in 2020 to 2021.

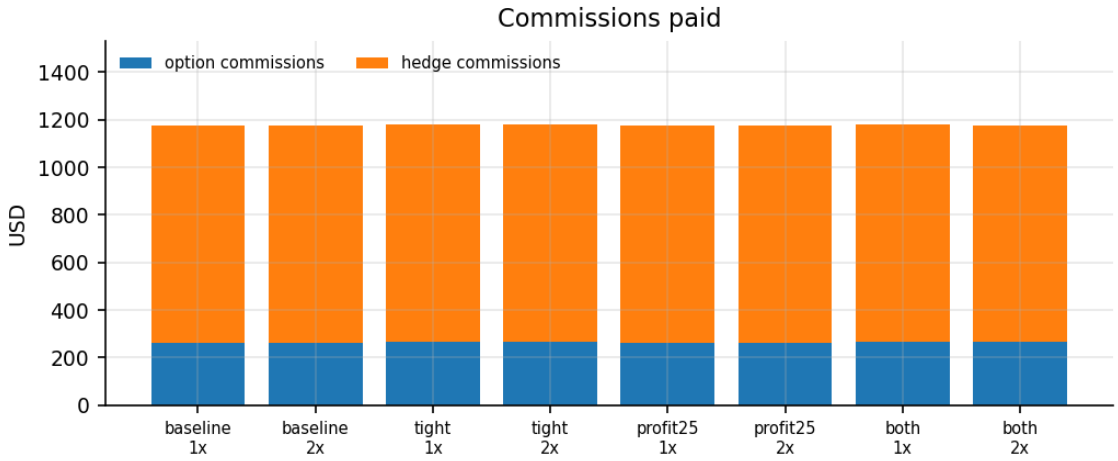


Figure 7: Commissions paid, by variant and spread

Commissions are the same for every variant (about $1,175, of which $910 to $914 is hedge orders at the $1 minimum). Fees don't explain the differences.

3 Analysis

The stops can't cut daily expected shortfall because of how they are checked. The worst 5% of days are 129 days in every run, and 127 of the 129 are the same days in `tight` and the baseline. The worst single day (−$468) is the same in all four variants. A stop checked once a day at 10:15 acts only after the move that triggers it. It can't stop the gap day that makes up the tail; it only closes the position the next morning. In this sample the baseline positions mostly recovered after such days: the worst short only reached −0.78x credit. Closing them at that point turned temporary losses into realized ones and slightly worsened both the tail and the P&L. The 20% ES cut that H1 needed would require removing losses on the tail days themselves, which a daily per-position stop can't do.

The 25% profit-take did what it promised on the trades it closed. Those positions were worth more when closed early than when held, and the first half (2014-2019) beat the baseline at both spreads. But closing a short early frees the book to open the next short, and in 2020-2021 those follow-on shorts lost more than the early exits had gained. That follow-on effect is a path effect from a few events (7 to 9 profit-takes), so the half-by-half result is close to noise. The pre-registered bar asked for a gain in both halves and over the full window, and it got neither the second half nor the full window.

`both` combines the two weaknesses: it is the worst variant at both spreads.

4 Verdict

H1 failed, H2 failed. The tighter stops made the daily tail slightly worse instead of cutting it by 20%, and at 2x spreads they also cost more than $300. The 25% profit-take lost to the baseline over the full window and in 2020-2024 at both spreads, although its tail was unchanged and it beat the baseline in 2014-2019. Neither change should go to `configs/paper.toml`. No hold-out run is needed.

5 Caveats

- **In-sample idea.** The idea for this experiment came from the baseline on the same window (no stop or profit-take ever fired there), so even a pass would only have been in-sample evidence.

- **Very few events.** 3 stops and 7 to 9 profit-takes in ten and a half years. The differences come from 2015, 2020-2021 and one profit-take in 2024.
- **Daily checks only.** Exits are checked once a day at 10:15, as in live trading. An intraday stop could behave differently. The replay snapshots support checks every 30 minutes (extra `[schedule] hedge` times, with spot implied from put-call parity), but that was not tested here.
- **One contract.** The $1 minimum per order is large relative to the position size (hedge commissions are about 78% of all fees).
- **Spread model.** It was fitted on about two weeks of calm 2026 live quotes; the 2x run is the stress check.
- **Missing days.** Month-end R2 days are missing; 3.3% of scheduled times were skipped in every run.

6 Trials

This experiment registered 6 trials (3 variants at 1x and 2x, from one run of `experiment.py`), plus the baseline's 2, for 8 in total. The baseline regression check that reproduced trial 1 was not registered. `tight` has exactly the same config as `cool1` in the parallel experiment 2026-10-16-stop-cooloff (0.5x credit, 0.25x premium, 1-day cool-off). The two gave identical numbers: all registry metrics, daily P&L and trade P&L match at 1x and 2x. The same test is therefore registered twice (4 trials) across the desk. The charts were rebuilt once with `--report-only` to fix an axis; that reads saved runs and adds no trial.